

Algebra First-Year Exam, January 2018, Part 2

Answer four problems. Write your answers clearly in complete English sentences. You may quote results (within reason) as long as you state them clearly.

1. Show that the ring

$$\mathbb{Z}[\omega] = \{a + b\omega \in \mathbb{C} \mid a, b \in \mathbb{Z}\}, \quad \omega = \frac{1 + i\sqrt{3}}{2}$$

is a Euclidean domain.

2. Let R be a commutative ring with identity. A subset $S \subseteq R$ is multiplicative if it is nonempty and is closed under multiplication. Show that if S is a multiplicative subset of R not containing 0, there is a prime ideal $\mathfrak{p} \subset R$ such that $\mathfrak{p} \cap S = \emptyset$. (First show using Zorn's lemma that the set of ideals $I \subset R$ such that $I \cap S = \emptyset$ has a maximal element. Then show, using an argument by contradiction that any such maximal element is prime).

3. Prove the following form of Eisenstein's criterion: Let R be an integral domain and $P \subset R$ a prime ideal. Suppose

$$f(X) = X^d + a_{d-1}X^{d-1} + \cdots + a_1X + a_0 \in R[X]$$

is a monic polynomial such that $a_i \in P$ for $i < d$ and $a_0 \notin P^2$. Then $f(X)$ is irreducible in $R[X]$.

4. Let R be an integral domain.

- (i) Define what it means for an element of R to be irreducible.
- (ii) Define what it means for an element of R to be prime.
- (iii) Show that a prime element is irreducible.

5. Suppose R is a commutative ring with identity, M and N are R -modules and $f : M \rightarrow N$ is an R -module homomorphism. Let $I \subset R$ be an ideal.

- (i) Show that there is a homomorphism $\bar{f} : M/IM \rightarrow N/IN$ such that

$$\bar{f}(m + IM) = f(m) + IN$$

for all $m \in M$.

- (ii) Show that if I is nilpotent (i.e. $I^k = 0$ for some $k > 0$) and $\bar{f}$ is surjective then f is surjective.

6. Find representatives of every similarity class of 6×6 matrices with rational coefficients whose minimal polynomial is $(x + 1)^2(x - 1)$.